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- Autonomous Anti-DDoS Network (A2D2)
- Video: Autonomous Anti-DDoS Network
- 13 min
- Reading: A2D2 Design and implementation
- 1h
- Practice Quiz: Practice Quiz 2.3.1
- 20 min
- Performance of CBQ/Rate Limit on DDoS Traffic Types
- Exam 2.3

● Congratulations! You passed!

Grade received 91.11% To pass 80% or higher

Practice Quiz. 2.3.1

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1. What are included in A2D2 DDoS Detection Evaluation Testbed?

1 / 1 point

Select all that apply.

Ready to review what you've learned before taking the quiz? I'm here to help.

☒ A Streaming subsystem for detecting the effectiveness of the related DDoS defenses.

● Submit your assignment

Correct

Correct. It was based on real player and server system for evaluating the impact of DDoS attacks and the effectiveness of various DDoS defense techniques.

Try again

☒ DDoS Detection and

Correct

Correct. It has firewall, Snort DDoS detection, and subnet flooding detection modules.

Your grade

91.11%

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We keep your highest score

☐ Email clients and servers.

Like

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☒ Stochastic DDoS traffic generating subsystem.

Correct

Correct. It generate DDoS attack packets.

2.

1 / 1 point

Iptables -A level2 -m -limit 100/s -j ACCEPT

Select all that apply.

☒ The above rule rate-limit the traffic on level2 chain to 100 packets per second.

Correct

Correct. That is the unit for the option --limit

☒ For any suspected sources such as 12.23.34.00/24 we can forward/join them to level2 chain with iptables -I Forward -s 12.23.34.00/24 -j level2 and subject them to the 100 packets/s restriction.

Correct

Correct. -j option will direct traffic that match the criteria (-s 12.23.34.00/24) to level 2 chain.

☒ We can use iptables -N level2 to announce/create a level2 chain.

Correct

Correct. It create a new chain with the parameter value.

☐ -limit 100/s is used to specify 100 kilo bits/sec restriction on bandwidth usage.

3. 0.0 / 1 point

A2D2 Adaptive Multi-Level Rate Limiting System.

Select all that apply.

☒ Persistent attack traffic last a long time will confirm the attacks and result in the blocking of the sources.

Correct

Correct. That is the intended purpose of the design.

☒ The multi-level limiting is implemented using iptables on firewall machine.

Correct

Correct. It is implemented by the combination of scripts and iptables firewall rules.

☒ The subnet flooding is implemented in as Snort preprocessor module.

Correct

Correct. Snort preprocessor allows us the compute statistics about the incoming traffic.

☒ The snort IDS is located in the same machine that runs the firewall.

Correct

Incorrect. The snort is located in a different machine. Subnet flooding detection results is sent in to the process running rate.pl which controls the firewall.

☐

The reason we design the adaptive multi-level rate limiting is that we are quite sure about the attacks patterns and related traffic thresholds.

4. Class-Based Queuing (CBQ) Implementation and Processing. 1 / 1 point

Select all that apply.

☒

CBQ then utilizes the tc command to setup different queues, each with their own allocated bandwidth and queue processing parameters.

Correct

Correct. A2D2 created a function that utilizes tc commands to make it easier for the user/programmer.

☒

CBQ first utilizes the iptables command to set the mark field value of the socket buffer which contains the packet with 1-4 in this case for specific protocol port #. It allows us to assign different media connections with different queues. Some queues are given preferred treatment with more resources.

Correct

Correct. By assigning proper bandwidth to each queue, we can isolate the impacts better.

☐

The other traffic that are not marked will be treated equally and routed out with the same bandwidth.

5. Commands for setting up CBQ 0.6666666666666666 / 1 point

Iptables -A FORWARD

-i tcp -s 0.0.0.0 -d 0.0.0.0 -p tcp -j MARK --set-mark 4

Select all that apply.

☒

This iptables command marks any packet with destination port 8080 with mark value 4.

Correct

Correct. We use --set-mark option for that purpose.